# Name: S.HEMAA
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# Experiment name: Experiment to detect outliers in a given data set

Description: Understand the procedure to identify the outliers in a given dataset

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#sample calculation for low range(lr) , upper range (ur),percentile
import numpy as np
array=np.random.randint(1,100,16) # randomly generate 16 numbers between 1 to 100
array
array([27, 50, 44, 6, 58, 61, 23, 86, 67, 20, 75, 7, 79, 61, 90, 54])
array.mean()
50.5
np.percentile(array,25)
26.0
np.percentile(array,50)
56.0
np.percentile(array,75)
69.0
np.percentile(array,100)
90.0
#outliers detection
def outDetection(array):
sorted(array)
Q1,Q3=np.percentile(array,[25,75])
IQR=Q3-Q1
lr=Q1-(1.5*IQR)
ur=Q3+(1.5*IQR)
return lr,ur
lr,ur=outDetection(array)
```

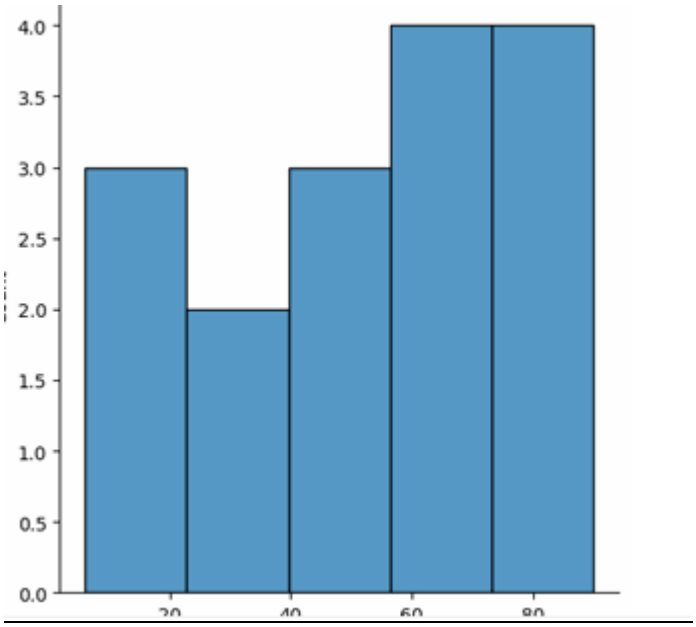
lr,ur

(-38.5, 133.5)

import seaborn as sns

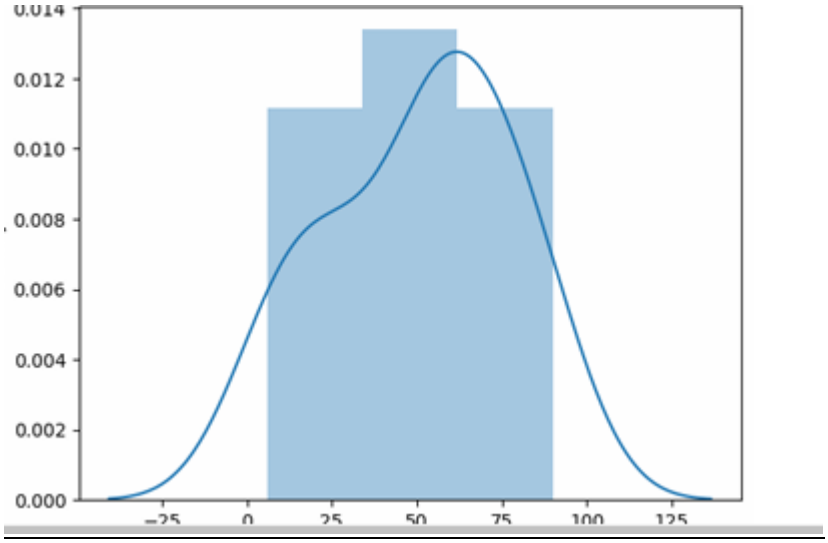
%matplotlib inline

sns.displot(array)




sns.distplot(array)

sns.distplot(array)

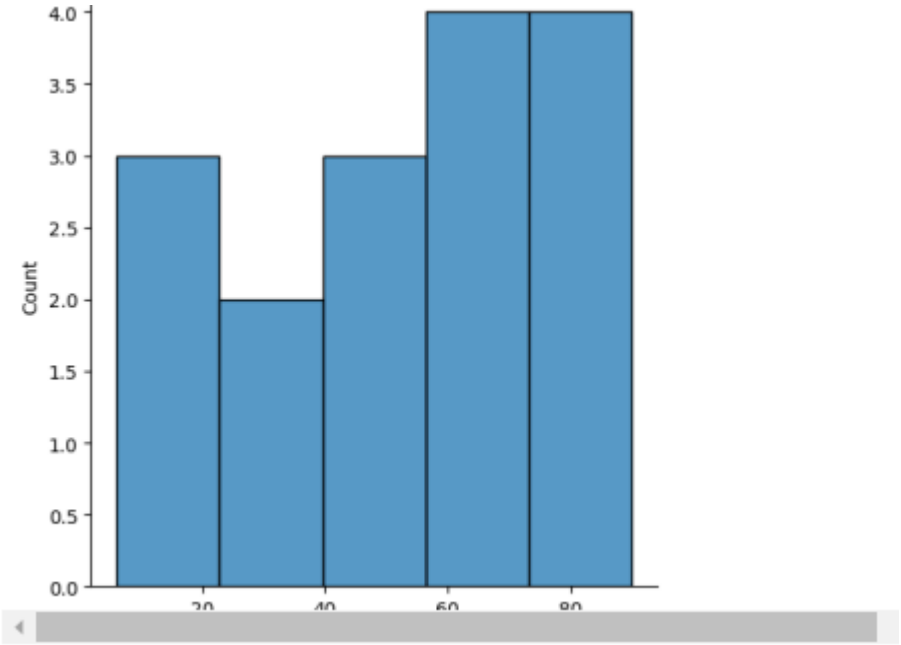



new_array=array[(array>lr) & (array<ur)]

new_array

array([27, 50, 44, 6, 58, 61, 23, 86, 67, 20, 75, 7, 79, 61, 90, 54])

<seaborn.axisgrid.FacetGrid at 0x78f2e09bb580>




lr1,ur1=outDetection(new_array)

lr1,ur1

(-38.5, 133.5)

final_array=new_array[(new_array>lr1) & (new_array<ur1)]

final_array

array([27, 50, 44, 6, 58, 61, 23, 86, 67, 20, 75, 7, 79, 61, 90, 54])

sns.distplot(final_array)

sns.distplot(final_array)

<Axes: ylabel='Density'>

